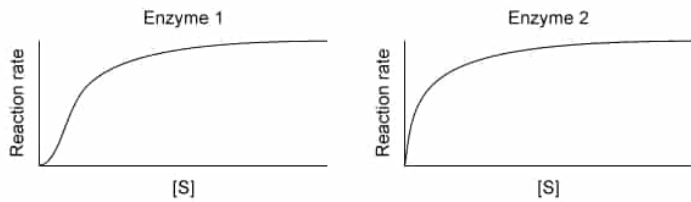


Researchers measured the activity of two enzymes that catalyze the same reaction, with results shown below. Which of the enzymes must have multiple active sites?



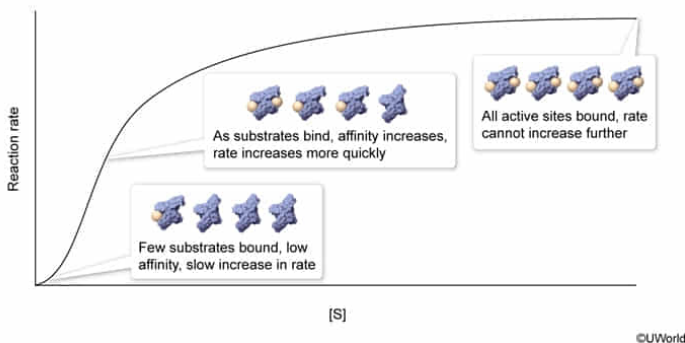
- ✔ ☒ A. Enzyme 1 only [74%]
☐ B. Enzyme 2 only [11%]
☐ C. Both enzymes [2%]
☐ D. Neither enzyme [11%]

Correct

73% answered correctly

Explanation:

Sigmoidal curves indicate enzymes with multiple, cooperative active sites



The rate of an enzyme-catalyzed reaction depends on the concentration of substrate. For enzymes with a single active site, the reaction rate increases steadily with substrate concentration until the active sites become saturated. Once saturation is achieved, adding more substrate does not increase the reaction rate. This takes the form of a **hyperbolic curve** when reaction rate is plotted as a function of substrate concentration.

Some enzymes have multiple active sites and can bind more than one substrate at a time. Many of these enzymes behave cooperatively such

that substrate binding at one active site alters the affinity of other active sites for the substrate. In **positive cooperativity**, binding at one active site increases the affinity, and therefore the reaction rate, of other active sites. This results in an "S"-shaped curve, called a **sigmoidal curve**, in which the rate increases slowly at first, increases more quickly as the affinity for substrate increases, and finally levels off as all active sites become saturated. Only enzymes with multiple active sites can exhibit cooperativity. Because it shows a sigmoidal curve, **enzyme 1 has positive cooperativity** and therefore **must have multiple active sites**.

(Choices B and C) Enzyme 2 has a hyperbolic curve, and therefore does not exhibit cooperativity. Although it is *possible* for a noncooperative enzyme to have multiple active sites, it is not required.

(Choice D) Enzyme 1 exhibits positive cooperativity, and therefore *must* have more than one active site.

Educational objective:

Enzymes that display positive cooperativity yield sigmoidal curves instead of hyperbolic curves when reaction rate is plotted against substrate concentration. Because cooperativity is a change in the affinity of one active site for its substrate when another active site is bound, any enzyme that exhibits cooperativity must have more than one active site.

Time spent:0 sec

Subject:
Biochemistry

QID:401526

System:
Enzymes